Yidi Miao

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Education

Carnegie Mellon University **Aug. 2023 – Present**
Doctor of Philosophy in Information Systems and Management *Pittsburgh, PA*

University of Michigan **Aug. 2021 – May 2023**
Masters of Science in Applied Math & Operations Research (Double Degrees) *Ann Arbor, MI*

Xi'an Jiaotong University **Aug. 2017 – Aug. 2021**
Bachelor of Engineering in Industrial Engineering *Xi'an, China*

Publication

- Robust Last-Mile Transportation Service Region and Dispatch Design Under Demand Ambiguity. (In Preparation)
- Beyond Single-Turn: A Survey on Multi-Turn Interactions with Large Language Models. (Submitted)
- Firm or Fickle? Evaluating Large Language Models Consistency in Sequential Interactions. ACL 2025.
- The PennSTART Safety Standards Project: Current Safety Standards and Test Track Designs for Connected and Autonomous Vehicles. USDOT National Transportation Library, 2024.
- Demand Learning and Supply Optimization for Last-Mile Transportation in Disadvantaged Neighborhoods. USDOT National Transportation Library, 2022.

Research

Transportation Service Region Design Under Demand Ambiguity

- **Objective:** Design an optimization framework to partition service regions and schedule vehicle dispatches for the last-mile transportation service under ambiguous demand distributions, reducing operational costs and complexities.
- **Methods:** Applied distributionally robust optimization (DRO) and reformulated a semi-infinite optimization problem into a finite semidefinite approximation, solvable in polynomial time to high precision using Gurobi/COPT.
- **Achievements:** Reproduced an analytic center cutting plane method to obtain the worst-case probability density function, and designed a tailored logic-based Benders decomposition algorithm for optimization.
- **Impact:** Developed a versatile and robust solution for last-mile transportation systems under distributional ambiguity.

Beyond single-turn: A Survey on Multi-Turn Interactions with Large Language Models

- **Objective:** Survey multi-turn interactions in large language models, categorize tasks, benchmarks, and improvement methods, and analyze challenges in context understanding, reasoning, fairness, and safety.

Evaluating Large Language Models Consistency in Sequential Interactions

- **Objective:** Investigate how large language models (LLMs) prioritize conversational flow over factual accuracy in multi-turn human-LLM interactions, and propose solutions to mitigate such bias.
- **Methods:** Designed a position-weighted consistency (PWC) score to evaluate LLMs' ability to maintain consistent responses and recovery patterns. Implemented confidence-aware response generation to enhance response stability.

Projects

Face Classification and Verification | *Python, PyTorch, AWS* **Apr. 2025**

- **Objective:** Build deep learning models for large-scale face recognition using VGGFace2 (9K+ identities, 18K+ pairs).
- **Achievements:** Applied contrastive learning with ResNet + ArcFace loss to learn discriminative embeddings, reducing EER to 3.3% and reaching 96.7% accuracy, demonstrating scalable face verification for identity management.

LoRA Fine-Tuning for Sentiment Analysis | *Python, PyTorch, AWS* **Mar. 2025**

- **Objective:** Explore parameter-efficient fine-tuning on BERT for five-class sentiment classification (SST-5).
- **Achievements:** Implemented LoRA over attention layers, delivering competitive SST-5 performance with 10× fewer parameters than full fine-tuning, converging faster and reducing overfitting.

Transformer for PJM Regulation Signal (Time Series) Forecasting | *Python, PyTorch, Scikit-learn* **Apr. 2022**

- Developed a Transformer for energy grid signal prediction, achieving <3% error on 500-point forecasts and improving computational efficiency by 25% over baseline LSTM, enabling real-time predictions to support energy system stability.

Comprehensive Studies of Water Ecology Management | *MySQL, R, Tableau* **Mar. – Jun. 2021**

- **Objective:** Evaluate long-term water quality trends and the effectiveness of remedial policies across 30 cities.
- **Methods:** Built a MySQL relational database with 1.3M+ records, applied advanced SQL functions to compute water quality indices, created Tableau dashboards, and performed causal inference (difference-in-difference) in R.
- **Impact:** Empirically showed stricter discharge standards and wastewater treatment investments improved water quality, while short-term bans had minimal effect, providing evidence for more effective policy design.

Experience

Cardinal Operations Co., Ltd. — **May - Aug. 2023**

Model improvement and algorithm deployment (intern) — *Beijing, China*

- Enhanced vehicle duty scheduling by reformulating it as a finite-horizon mixed-integer coloring problem, cutting optimization time by 43% and enabling real-time allocation decisions for client fleets.
- Optimized subway scheduling with time-window constraints using Gurobi/COPT, reducing decision variables by 94% and improving computational efficiency to support more reliable transit planning.

Ernst & Young — **July - Aug. 2020**

Risk management and internal control (intern) — *Beijing, China*

- Collaborated with seniors on automobile industry research using SWOT analysis, and created client-facing slides.
- Organized client interviews, ensuring project execution and fostering partnerships through relationship management.

Technical Skills

- **ML/AI** PyTorch, Scikit-learn, HuggingFace, LangFlow, Spark
- **Optimization** Gurobi, COPT, Cplex, CVX, GAMS
- **Data Tools** Git, Bash, AWS (EC2, S3, SageMaker), Tableau, LaTeX
- **Languages** Python, SQL, R, Matlab, Java, C/C++

Courses

- **Teaching Assistant**
  - IOE 510 *Linear Programming I* with Prof. Jon Lee at UMich IOE
  - 94-867 *Decision Analytics for Business and Policy* with Prof. Peter Zhang at CMU Heinz
  - 95-760 *Decision Making Under Uncertainty* with Prof. Peter Zhang at CMU Heinz
  - 95-760 *Decision Making under Uncertainty* with Prof. David Choi at CMU Heinz
  - 95-865 *Unstructured Data Analysis* with Prof. George Chen at CMU Heinz
- **Coursework**
  - EECS 61A *Structure and Interpretation of Computer Programs* (UC Berkeley)
  - STAT 150 *Stochastic Processes* (UC Berkeley)
  - IOE 511 *Continuous Optimization Methods* (UMich)
  - IOE 512 *Dynamic Programming* (UMich)
  - IOE 516 *Stochastic Processes II* (UMich)
  - IOE 610 *Linear Programming II* (UMich)
  - IOE 611 *Nonlinear Programming* (UMich)
  - IOE 618 *Stochastic Optimization* (UMich)
  - EECS 545 *Machine Learning* (UMich)
  - 10-605 *Machine Learning with Large Dataset* (CMU)
  - 10-715 *Advanced Introduction to Machine Learning* (CMU)
  - 11-711 *Advanced Natural Language Processing* (CMU)
  - 11-785 *Introduction to Deep Learning* (CMU)
  - 15-513 *Introduction to Computer Systems* (CMU)
  - 36-705 *Intermediate Statistics* (CMU)